

WORKED EXAMPLE

RProp follows derivative signs and remembers state

Use the simplified rollback, shrink, retry convention in Lecture4 PDF page71.

$$E(w) = \frac{1}{2}(w - 1)^2, \quad g(w) = w - 1$$

$$w = 0, \quad \text{prevD} = -1, \quad s = 0.6$$

Store three quantities: accepted parameter w , its derivative prevD , and a positive step magnitude s . Unlike ordinary gradient descent, the derivative magnitude does not directly set the move length.

$$\Delta = \text{sign}(\text{prevD})s, \quad w_{\text{trial}} = w - \Delta$$

Sign is +1 for a positive derivative and -1 for a negative derivative. Delta is the signed quantity subtracted. A negative Delta therefore moves w upward.

Attempt 1 uses the initial magnitude 0.6 immediately. There is no growth step before that first trial.

WORKED EXAMPLE

Decide which state is carried to the next attempt

A trial is not an accepted parameter until the sign check passes.

$$D = g(w_{trial}), \quad \text{sign test: } prevD \cdot D$$

Same nonzero signs: accept the trial, replace $prevD$ with D , and grow the positive magnitude by 1.2, capped at 1.

$$s_{next} = \min(1.2s, 1)$$

Opposite signs: undo the trial, keep the old accepted w and old $prevD$, and shrink the magnitude by 0.5, floored at 0.01.

$$w_{restored} = w_{trial} + \Delta, \quad s_{next} = \max(0.5s, 0.01)$$

Here an exact zero derivative means stop at that point. Bounds apply to the positive magnitude before attaching its sign; this clarifies the signed min/max shorthand in the slide.

WORKED EXAMPLE

attempt 1: accept and grow

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0, \quad prevD = -1, \quad s = 0.6$$

$$\Delta = (-1)(0.6) = -0.6, \quad w_{trial} = 0 - (-0.6) = 0.6$$

$$D = 0.6 - 1 = -0.4, \quad prevD \cdot D = (-1)(-0.4) = 0.4$$

$$s_{next} = \min(1.2(0.6), 1) = 0.72$$

$$(w, prevD, s)_{next} = (0.6, -0.4, 0.72)$$

The sign product is positive. Accept this trial and store its derivative; the larger magnitude is for the next attempt.

WORKED EXAMPLE

attempt 2: rollback and shrink

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0.6, \quad \text{prevD} = -0.4, \quad s = 0.72$$

$$\Delta = (-1)(0.72) = -0.72, \quad w_{\text{trial}} = 0.6 - (-0.72) = 1.32$$

$$D = 1.32 - 1 = 0.32, \quad \text{prevD} \cdot D = (-0.4)(0.32) = -0.128$$

$$w_{\text{restored}} = 1.32 + (-0.72) = 0.6$$

$$s_{\text{next}} = \max(0.5(0.72), 0.01) = 0.36$$

$$(w, \text{prevD}, s)_{\text{next}} = (0.6, -0.4, 0.36)$$

The sign product is negative. Reject this trial; its derivative does not replace prevD. Retry from the restored accepted parameter.

WORKED EXAMPLE

attempt 3: accept and grow

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0.6, \quad prevD = -0.4, \quad s = 0.36$$

$$\Delta = (-1)(0.36) = -0.36, \quad w_{trial} = 0.6 - (-0.36) = 0.96$$

$$D = 0.96 - 1 = -0.04, \quad prevD \cdot D = (-0.4)(-0.04) = 0.016$$

$$s_{next} = \min(1.2(0.36), 1) = 0.432$$

$$(w, prevD, s)_{next} = (0.96, -0.04, 0.432)$$

The sign product is positive. Accept this trial and store its derivative; the larger magnitude is for the next attempt.

WORKED EXAMPLE

Why rollback is not a loss-improvement test

The rejected trial can have lower loss than the accepted point.

$$E(0.6) = \frac{1}{2}(0.6 - 1)^2 = 0.08$$

$$E(1.32) = \frac{1}{2}(1.32 - 1)^2 = 0.0512$$

Even though 0.0512 is lower than 0.08, attempt2 is rejected because the derivative changes from -0.4 to +0.32. This simplified rule uses the sign change, not a loss comparison.

$$\text{retry : } 0.6 - (-0.36) = 0.96$$

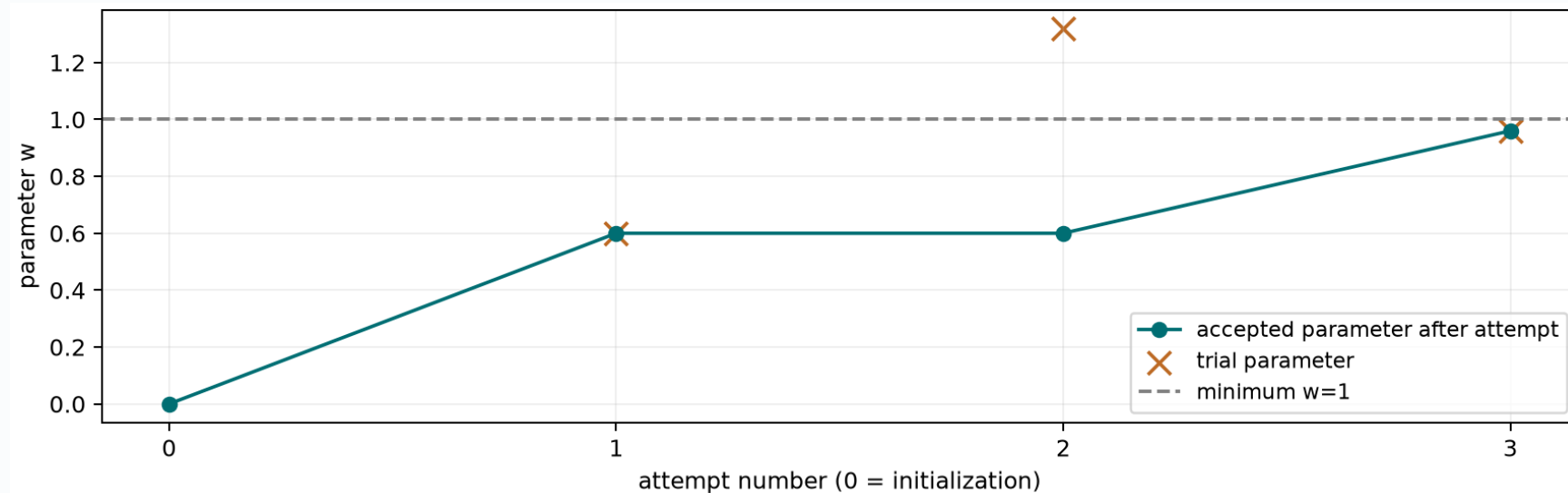
$$E(0.96) = \frac{1}{2}(-0.04)^2 = 0.0008$$

Accepted parameters after the attempts are 0.6, 0.6, 0.96. Rejected attempts count as work, but do not advance the accepted parameter.

WORKED EXAMPLE

Separate attempted points from accepted history

Crosses mark trials; a separate cross marks a rejected trial.



Attempt2 visits 1.32, then returns to the accepted 0.6. Attempt3 starts from 0.6 with the smaller magnitude, not from the rejected point.

WORKED EXAMPLE

Clamp a magnitude, then attach its direction

A signed quantity and a positive magnitude are different objects.

$$s = 0.9 : \quad s_{grow} = \min(1.2(0.9), 1) = \min(1.08, 1) = 1$$

$$s = 0.015 : \quad s_{shrink} = \max(0.5(0.015), 0.01) = 0.01$$

$$prevD < 0, \quad s = 1 \Rightarrow \Delta = -1$$

$$prevD > 0, \quad s = 0.01 \Rightarrow \Delta = +0.01$$

The lower and upper bounds protect the step length. Applying min or max directly to a negative signed step would give the wrong bound behavior.

The lecture gives separate state for each parameter. In a network, each coordinate has its own derivative sign and step magnitude; they are not one shared learning rate.

INDEPENDENT PRACTICE

Your turn: four attempts with two reversals

Keep the same growth, shrink and bounds, but use a fresh quadratic.

$$E(w) = \frac{1}{2}w^2, \quad g(w) = w$$

$$w = 1, \quad \text{prevD} = 1, \quad s = 0.7$$

Trace four attempts. For each, write the signed step, trial, trial derivative, sign product, decision, and full next state (w,prevD,s).

Explain why a rejected negative derivative must not become prevD. Separately compute the bounded growth of 0.9 and bounded shrink of 0.015.

The following pages show every attempted move. This practice follows the stated lecture variant; other RProp variants can use different reversal rules.

WORKED ANSWER

Practice: attempt 1: accept and grow

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 1, \quad prevD = 1, \quad s = 0.7$$

$$\Delta = (1)(0.7) = 0.7, \quad w_{trial} = 1 - (0.7) = 0.3$$

$$D = 0.3 - 0 = 0.3, \quad prevD \cdot D = (1)(0.3) = 0.3$$

$$s_{next} = \min(1.2(0.7), 1) = 0.84$$

$$(w, prevD, s)_{next} = (0.3, 0.3, 0.84)$$

The sign product is positive. Accept this trial and store its derivative; the larger magnitude is for the next attempt.

WORKED ANSWER

Practice: attempt 2: rollback and shrink

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0.3, \quad \text{prevD} = 0.3, \quad s = 0.84$$

$$\Delta = (1)(0.84) = 0.84, \quad w_{\text{trial}} = 0.3 - (0.84) = -0.54$$

$$D = -0.54 - 0 = -0.54, \quad \text{prevD} \cdot D = (0.3)(-0.54) = -0.162$$

$$w_{\text{restored}} = -0.54 + (0.84) = 0.3$$

$$s_{\text{next}} = \max(0.5(0.84), 0.01) = 0.42$$

$$(w, \text{prevD}, s)_{\text{next}} = (0.3, 0.3, 0.42)$$

The sign product is negative. Reject this trial; its derivative does not replace prevD. Retry from the restored accepted parameter.

WORKED ANSWER

Practice: attempt 3: rollback and shrink

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0.3, \quad \text{prevD} = 0.3, \quad s = 0.42$$

$$\Delta = (1)(0.42) = 0.42, \quad w_{\text{trial}} = 0.3 - (0.42) = -0.12$$

$$D = -0.12 - 0 = -0.12, \quad \text{prevD} \cdot D = (0.3)(-0.12) = -0.036$$

$$w_{\text{restored}} = -0.12 + (0.42) = 0.3$$

$$s_{\text{next}} = \max(0.5(0.42), 0.01) = 0.21$$

$$(w, \text{prevD}, s)_{\text{next}} = (0.3, 0.3, 0.21)$$

The sign product is negative. Reject this trial; its derivative does not replace prevD. Retry from the restored accepted parameter.

WORKED ANSWER

Practice: attempt 4: accept and grow

Read the starting state first; use the next magnitude only on the next attempt.

$$w = 0.3, \quad prevD = 0.3, \quad s = 0.21$$

$$\Delta = (1)(0.21) = 0.21, \quad w_{trial} = 0.3 - (0.21) = 0.09$$

$$D = 0.09 - 0 = 0.09, \quad prevD \cdot D = (0.3)(0.09) = 0.027$$

$$s_{next} = \min(1.2(0.21), 1) = 0.252$$

$$(w, prevD, s)_{next} = (0.09, 0.09, 0.252)$$

The sign product is positive. Accept this trial and store its derivative; the larger magnitude is for the next attempt.

WORKED ANSWER

Practice checkpoint: track accepted state

Rejected moves keep the same accepted position and derivative.

attempt	trial	decision	accepted w	prevD	next s
1	0.3	accept	0.3	0.3	0.84
2	-0.54	reject	0.3	0.3	0.42
3	-0.12	reject	0.3	0.3	0.21
4	0.09	accept	0.09	0.09	0.252

The bound answers are 1 and 0.01, as worked earlier. There are four attempts but only two accepted moves in this practice.

Common errors: growing before the first trial, replacing prevD after rejection, continuing from a rejected parameter, using loss improvement as the sign test, or confusing the signed Delta with its positive magnitude.